DATA ANALYSIS

Data analysis is a method in which data is collected and organized so that one can derive helpful information from it. In other words, the main purpose of data analysis is to look at what the data is trying to tell us. For example, what does the data show or do? What does the data not show or do?

SCALES OF MEASUREMENT

Nominal Scales

A nominal scale is a classification system that places people, objects, or other entities into mutually exclusive categories. A variable measured using a nominal scale is a classification variable that indicates the group to which each subject belongs. The examples of classification variables provided earlier (Gender and Race) also serve as examples of nominal variables. They tell us to which group a subject belongs, but they do not provide any quantitative information about the subjects.

Ordinal Scales

Values on an ordinal scale represent the rank order of the subjects with respect to the variable being assessed.

Interval Scales

With an interval scale, equal differences between scale values do have equal quantitative meaning. For this reason, an interval scale provides more quantitative information than the ordinal scale. A good example of an interval scale is the Fahrenheit degree scale used to measure temperature. With the Fahrenheit scale, the difference between 70 degrees and 75 degrees is equal to the difference between 80 degrees and 85 degrees: The units of measurement are equal throughout the full range of the scale.

Ratio Scales

Ratio scales are similar to interval scales in that equal differences between scale values have equal quantitative meaning. However, ratio scales also have a true zero point, which gives them an additional property. With ratio scales, it is possible to make meaningful

statements about the ratios between scale values. For example, the system of inches used with a common ruler is an example of a ratio scale. There is a true zero point because zero inches does in fact indicate a complete absence of length. With this scale, it is possible to make meaningful statements about ratios. It is appropriate to say that an object four inches long is twice as long as an object two inches long. Age, as measured in years, is also on a ratio scale—a 10-year-old house is twice as old as a 5-year-old house. Notice that it is not possible to make these statements about ratios with the interval-level variables discussed above. You would not say that a person with an IQ of 160 is twice as intelligent as a person with an IQ of 80 because there is no true zero point on the IQ scale.

QUANTITATIVE DATA ANALYSIS

The first step in data analysis is to describe, or summarize, the data using descriptive statistics. Descriptive statistics permit the researcher to meaningfully describe many pieces of data with a few indices. If such indices are calculated for a sample drawn from a population, the resulting values are referred to as statistics; if they are calculated for an entire population, they are referred to as parameters.

a statistic is a quantitative index that describes performance of a sample or samples, and a parameter is a quantitative index describing the performance of a population.

The major types of **DESCRIPTIVE STATISTICS** are measures of central tendency, measures of variability, measures of relative position, and measures of relationship. Measures of central tendency are used to determine the typical or average score of a group of scores. Measures of variability indicate how spread out a group of scores are. Measures of relative position describe a participant's performance compared to the performance of all other participants. Measures of relationship indicate the degree to which two sets of scores are related (remember correlation?)

Before actually calculating any of these measures, it is often useful to present the data in graphic form.

1. **Measures of central tendency**

Measures of central tendency provide a convenient way of describing a set of data with a single number. The number resulting from computation of a measure of central tendency represents the average or typical score attained by a group of subjects. The three most frequently encountered indices of central tendency are the mode, the median, and the mean. Each of these indices is used with a different scale of measurement: the mode is appropriate for describing nominal data, the median for describing ordinal data, and the mean for describing interval or ratio data.

2. **Measures of variability**

Although measures of central tendency are very useful statistics for describing a set of data, they are not sufficient. Two sets of data that are very different can have identical means or medians. As an example, consider the following sets of data:

set A: 79 79 79 80 81 81 81

set B: 50 60 70 80 90 100 110

The mean of both sets of scores is 80 and the median of both is 80, but set A is very different from set B. In set A the scores are all very close together and clustered around the mean. In set B the scores are much more spread out; in other words, there is much more variation or variability in set B. Thus, there is a need for a measure that indicates how spread out the scores are, that is, how much variability there is. There are a number of descriptive statistics that serve this purpose, and they are referred to as measures of variability**.** The three most frequently encountered are the range, the quartile deviation, and the standa**rd deviation.**

3. **Measures of relative position**

Measures of relative position indicate where a score is in relation to all other scores in the **d**istribution. In other words, measures of relative position permi**t one to express how well an in**dividual has performed as compared to all other individuals in the sample who have been

measured on the same variable. A major advantage of such measures is that they make it possible to compare the performance of an individual on two or more different tests.

Percentile Ranks

A percentile rank indicates the percentage of scores that fall at or below a given score. Percentiles are appropriate for data representing an ordinal scale, although they are mainly computed for interval data. The median of a set of scores corresponds to the 50th percentile, which makes sense since the median is the middle point and therefore the point below which are 50% of the scores. While percentile ranks are not used very often in research studies, they are frequently used in the public schools to report test results of students in a form that is understandable to most audiences.

Standard Scores

Basically, a standard score is a derived score that expresses how far a given raw score is from some reference **point, typically the mean, in** terms of standard deviation units. A standard score is a measure of relative position which is appropriate when the test data represent an interval or ratio scale of measurement. The most commonly reported and used standard scores are 2 scores, T score**s (or Z scores), and stanines.** Standard scores allow scores from different tests to be compared on a common scale and, unlike percentiles, permit valid mathematical operations such as a**verages to be computed.** A z score is the most basic standard score. It expresses how far a score is from the mean in terms of standard deviation units. A score that is exactly "on" the mean corresponds to a z score of 0.

A T score (also called a Z score) is nothing more than a z score expressed in a different form. To transform a z score to a T score, you simply multiply the z score by 10 and add 50.

Stanines. Stanines are standard scores that divide a distribution into nine parts. Stanine (short for "standard nine") equivalencies are derived using the formula 22 + 5 and rounding resulting values to the nearest whole number. Stanines 2 through 8 each represent % SD of the distribution; **stanines 1 and 9 inclu**de the remainder.

4. **Measures of relationship**

Degree of relationship is expressed as a correlation coefficient, which is computed using two sets of scores from a single group of participants. The correlation coefficient provides an estimate just how related two variables are. If two variables are highly related, a correlation coefficient near +1.00 (or - 1.00) will be obtained; if two variables are not related, a coefficient near .00 will be obtained. There are a number of different methods of computing a correlation coefficient; **w**hich one is appropriate depends on the scale of measurement represented by the data. The two most frequently used correlational analyses are the rank difference correlation coefficient, usually referred to as the **Spearm**an rho, and the product moment correlation coefficie**nt, usually referred to as the Pearson** r

INFERENTIAL STATISTICS

Inferential statistics are the statistical procedures that are used to reach conclusions about **a**ssociations between variables. inferential statistical analysis involves using information from a sample to make inferences, or estimates, about the population. For example, assume that you need to know how much insurance is sold by the average agent in the company. Suppose it is impossible (or very difficult) to obtain the necessary information from all 10,000 agents and then calculate a mean. An alternative is to draw a random (and ideally representative) sample of 100 agents and determine the average amount sold by this subset. If this group of 100 sold an average of $179,322 worth of policies last year, then your best guess of the amount of insurance sold by all 10,000 agents would be $179,322. You have used characteristics of the sample to make inferences about characteristics of the population. Using some simple statistical procedures, you can even compute confidence intervals around the estimate, which allows you to make statements such as “There is a 95% chance that the actual population mean lies somewhere between $172,994 and $185,650.” This is the real value of inferential statistical procedures—they allow you to review information obtained from a relatively small sample and then to make inferences about a population**.**

Qualitative data analysis

1 Content Analysis:

Content analysis is the procedure for the categorization of verbal or **behavioral** data for the purpose of classification, summarization and tabulation. Content analysis can be done on two levels: Descriptive: What is the data? Interpretative: what was meant by the data?

2 Narrative Analysis

Narratives are transcribed experiences. Every interview/observation has narrative aspect. The researcher has to sort-out and reflect up on them, enhance them and present them in a revised shape to the reader. The core activity in narrative analysis is to reformulate stories presented by people in different contexts and based on their different experiences.

3 Discourse Analysis

This is a method of analyzing a naturally occurring talk (spoken interaction) and all types of written texts. It focuses on how people express themselves verbally in their everyday social life i.e. how language is used in everyday situations?

- Sometimes people express themselves in a simple and straightforward way
- Sometimes people express themselves vaguely and indirectly
- Analyst must refer to the context when interpreting the message because the same phenomenon can be described in a number of different ways depending on **context**

4 Framework Analysis

Framework is an approach to qualitative data analysis that offers researchers a systematic structure to manage, analyze and identify themes, and is particularly useful with large volumes of text. It was developed by Ritchie and Spencer in the 1980s at the UK's National Centre for Social Research for use in policy development research.

Five main stages in framework analysis

Familiarization

This early stage is for the researchers to get familiarized with the data and sensitized to early themes. It encourages the research to see the individual differences inherent in transcripts that can sometimes get lost when coding begins. The process of sensitization to these individual

differences also enables the researcher to better identify within- and between-participant differences.

Identifying a thematic framework

This stage of framework analysis is commonly referred to as ‘coding’ in other qualitative methodologies. This principally involves identifying key themes, issues or discussion points embedded in the transcript. These are delineated and assigned a ‘code’ or a name that best captures the essence of the theme or issue identified. In framework analysis, ‘a priori issues’ questions can form the basis of the themes.

Indexing

Indexing refers to the process of numerically annotating transcripts in order to identify consistencies, which then go on to develop the coding framework.

Charting

Framework analysis describes this stage as a process of rearranging the data and thematic framework to create order, not dissimilar to the iterative principle of grounded theory.

Mapping and interpretation

Mapping and interpreting essentially are ways of representing pictorially or graphically all of the themes and investigating how each of the themes relates to each other.

Example:

Main Theme	Minor categories	Explanation
Access to MIS	Collecting and inputting data	Participants spoke about data collection as a predominantly paper based exercise, although this was seen as pragmatic and cost driven, rather than ideal.

	Setting and time of MIS access	Data entry was usually an office based procedure, whereas review and analysis might rely on remote access. Modern technology facilitated multiple simultaneous use
	Reasons for stakeholder access	Access for data input, administration, monitoring and evaluation was described
	Support and training in MIS use	Both formal and informal training were reported with the need for training to be an ongoing process.
	Use of the data	Use of MIS data in administration, monitoring the progress of individual participants and monitoring the progress of the entire project were described.
Data and its function	Confidence in the data	Data quality was a frequent theme that included accurate and secure data with consistency of MIS use, safeguards and 'back –up'
	Data processing	Data cleaning, extraction, analysis and linkage were explored
	Use of the data	Use of MIS data in administration, monitoring the progress of individual participants and monitoring the progress of the entire project were described.

5) Grounded Theory:

This theory starts with an examination of a single case from a ‘pre-defined’ population in order to formulate a general statement (concept or a hypothesis) about a population. Afterwards the analyst examines another case to see whether the hypothesis fits the statement. If it does, a further case is selected but if it doesn’t fit there are two options:

Either the statement is changed to fit both cases or the definition of the population is changed in such a way that the case is no longer a member of the newly defined population.

Then another case is selected and the process continues. In such a way one should be able to arrive at a statement that fits all cases of a population-as-defined. This method is only for limited set of analytic problems: those that can be solved with some general overall statement .